MODULE I

Memory organization: Memory hierarchy – Main memory – Auxiliary memory –Associative memory – Cache memory – Virtual memory, Input-output organization: Peripheral Devices – I/O interface – Asynchronous data transfer-DMA-Input/output processor (IOP)

MODULE II

CPU: Register and stack organization – Instruction formats – Addressing modes– Data transfer and manipulation Operating Systems -OS structure – batch processing - multiprogramming - time sharing - OS operations. Operating system services – User OS interface, System calls – types, System programs, OS structure – simple – layered, virtual machine – introduction only, system boot.

MODULE III

Introduction to programming languages – types of programming languages – high level – assembly language – low level language – Compilers and interpreters. Problem solving concepts– flow charts and algorithms – Features of C language. Introduction to C programming – variables and arithmetic expressions- Formatted output- printf()